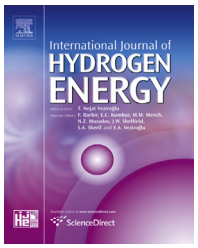
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Concentration polarization distribution along Pd-based membrane reactors: A modelling approach applied to Water-Gas Shift

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ABSTRACT

In this work, an improved membrane characterization, computational fluid dynamic (CFD) simulations of inert particle beds, and simulations of a membrane reactor for Water Gas Shift (WGS) are coupled together to evaluate the concentration polarization distribution along Pd-based membrane reactors. To perform this investigation, first a 3.6 μm-thick $Pd_{75}Ag_{25}$-membrane is characterized using experimental data from the literature as input to an improved approach allowing the non-ideality of internal diffusion to be quantitatively evaluated as a function of hydrogen partial pressure. Then, CFD simulations of a particle bed composed of regular arrays of mono-disperse spherical particles are carried out, showing that the velocity field between particles and membranes contributes to enhance the mass transfer towards the membrane surface. Finally, after validating a 1D-1D reactor model by literature experimental data in terms of both conversion and recovery index vs. feed pressure, simulations of concentration polarization coefficient (CPC) profiles along the reactor are performed. These profiles allow a clearly identification of the reactor zones where polarization is higher, providing useful information to minimize the polarization influence on reactor performances.

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Introduction

Membrane-assisted reactors offer a unique opportunity to combine reaction and separation in a single device, reducing the costs in terms of plant volume, better raw materials exploitation and energy efficiency [1]. In designing a membrane device, one of the most important aspects is the fluid dynamic condition, which is the key-issue to act on to reduce the mass transfer resistance outside membrane, which is named as "*concentration polarization*" on the feed side.

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Concentration polarization is an important and often underestimated phenomenon affecting the membrane processes. For this reason, an approach able to systematically take into account this phenomenon is needed for an efficient design of membrane-assisted devices.

To reduce the influence of such a phenomenon, due to the resistance of the less permeating species to the flux of the more permeating ones [2–18], several design solutions were proposed, such as micro-structured membrane modules [19], fluidized bed reactors/permeators [20] and devices based on turbulence-promoted flow [5–10,21–24].

In general, the effect of concentration polarization is gradually stronger for increasing membrane selectivity and permeance. Therefore, it is quite important, for example, in hydrogen permeation through Pd-based membranes, which show an infinite selectivity with respect to the other species by letting only hydrogen pass through.

Concentration polarization quantifying is not an easy issue, as it is usually difficult to measure the external mass transfer resistance directly from experimental data. More often, simulation of the considered membrane device is performed by using an appropriate transport model matching the measurable device performance, basically conversion and recovery. This is because the actual concentration of the species in the neighbourhood of membrane surface is generally unknown, especially in systems where the flow is interrupted by obstacles (like catalyst particles or tube bundles) enhancing the external mass transfer.

A possible approach to simulation of these types of systems could be to use techniques of computational fluid dynamics (CFD), implementing the Navier–Stokes equations along with the appropriate constitutive equations and boundary conditions with a brute-force approach to reproduce at best the real fluid dynamics behaviour of the flow in the device taking into account the heterogeneity of the system (i.e., the presence of catalyst particles and obstacles). If successful, such a simulation would provide us with all information needed to evaluate the external mass transfer resistance in each axial position of membrane surface.

To do that, however, there are several issues to deal with. In fact, even in fully-developed laminar conditions, obstacles can induce local turbulences generating vortices, whose presence is not easy to take into account by CFD. This could make the actual fluid dynamic behaviour quite different from the simulated one, this resulting in over or under estimation of the external mass transfer resistance and, thus, of the overall device performance. To partially address this issue, appropriate turbulence models could be used locally [25–30].

However, the necessary computational time could make the simulation prohibitive. Therefore, in most cases simplified constitutive mass and heat transfer models based on semi-empirical but well-established correlations are used within mono-dimensional conservation equations.

In this context, the present work uses a systematic approach to evaluate the concentration polarization coefficient (CPC) profiles in catalytic-bed membrane reactors. For this purpose, a high-performance Pd-based membrane of literature [31] is characterized by an improved approach to evaluate the intrinsic permeation parameters in terms of permeability and non-ideal contribution of internal diffusion. Then, the so-evaluated membrane properties are used to perform computational fluid dynamics (CFD) simulations considering a particle bed with two different values of particle size.

As well, the same membrane properties are input into a 1D-1D transport reactor model to simulate the behaviour of a membrane reactor for Water-Gas Shift, including the radial hydrogen concentration drop owing to the presence of the non-permeating species involved in the reaction. The model is validated with experimental reaction data from the literature in terms of both CO conversion and H_2 recovery [31].

Modelling and simulation approach

Model of membrane reactors

Based on the approach adopted in this work, simulations of concentration polarization in particle-bed membrane reactors is the result of coupling between a permeation model of the hydrogen transport through membrane and a model of reactor.

As for the permeation model, Fig. 1 shows the two steps considered: 1) the external mass transfer from the bulk of the particles bed towards the membrane surface and 2) the internal diffusion of hydrogen through the Pd-based membrane selective (dense) layer. Since in the permeate side pure hydrogen is considered, no transport equations are needed (flat radial pressure profile).

As for the first step, an already developed and validated multicomponent-based permeation model is improved [2,32,33] to account for the presence of the particle (catalyst) bed. Such improvements consist in using both an effective diffusivity corrected by the $\varepsilon_{Bed}/\tau_{Bed}$ ratio (=0.51/1.7) and the effective inter-particle velocity to estimate the mass transfer film thickness [34]. Concerning the diffusion in the Pd-layer, it is modelled by the irreversible thermodynamics theory, for which the hydrogen chemical potential gradient is the driving force, considering the non-ideality of diffusion coefficient. The required membrane properties are evaluated by applying

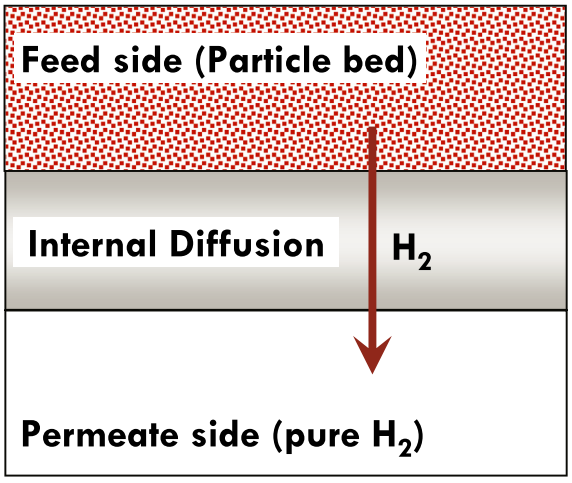


Fig. 1 – Sketch of the permeation model used to evaluate the permeating flux through Pd-based membrane.

the improved approach of Hara and co-workers [35,36] to a 3.6 um-thick $Pd_{75}Ag_{25}$-membrane from the literature [31].

As for the transport model of the whole reactor (non-isothermal), a mono-dimensional model previously used for methane steam reforming [37–39] is here modified to account for the radial profiles by keeping the plug-flow hypothesis along the axial direction but using the above-mentioned multicomponent-based membrane permeation model along the radial direction at each axial position. In this sense, the whole model can be named as "1D + 1D", as the transport phenomena along the radial and axial direction are solved in combination. The sketch of the simulated system considering five tubular 5 μm-thick membranes under the operating conditions reported in Table 1 is described in Fig. 2. The kinetic model used for simulation is that developed by Keiski et al. (1996) [40] for high temperature Water Gas Shift, here reported for more clarity (Eq. (1)):

$$-r_{CO} = k_{CO} P_{CO}^{1.1} P_{H_2O}^{0.53} \left(1 - \frac{P_{CO_2} P_{H_2}}{P_{CO} P_{H_2O}} \frac{1}{K_{Eq}}\right)$$
$$k_{CO} = 2.2 \times 10^{11} e^{\left(\frac{-11426.5}{T[K]}\right)} \quad (1)$$
$$K_{Eq} = \exp\left(-4.33 + \frac{4577.3}{T[K]}\right)$$

CFD simulation for concentration polarization

In parallel with using mono-dimensional models, the influence of concentration polarization is preliminary investigated here also by computational fluid dynamics (CFD) techniques applied to beds of particles. For this purpose, the concentration polarization behaviour is analyzed in an inert 20 cm-long tubular particle-bed (ID = 2.54 cm) equipped with a single 5 μm-thick supported membrane at 360 °C.

Such an investigation is carried out to give a preliminary idea of the polarization level in beds of particles, whose fluid dynamics is similar to that in catalytic chemical reactors.

For this purpose, the Navier–Stokes equations coupled with the mass conservation ones are implemented in Comsol Multiphysics® considering a H_2/N_2 binary mixture with a hydrogen feed composition of 98.6% flowing along a particle-bed of regularly-distributed spherical particles. The use of such a high value of hydrogen composition is made to consider a wider range of compositions, starting from a maximum composition close to the unity – corresponding to values typical of high-purification processes (where concentration polarization level close to zero) – for then decreasing it down to lower values typical of more standard separation processes.

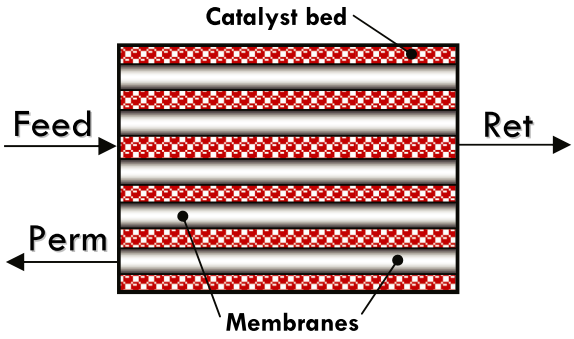


Fig. 2 – Configuration considered in the reactor simulation. Five tubular membranes equally distributed over the cross-sectional area of reactor.

For simulation, a 2-D axial-symmetry geometry is setup (see Fig. 6a in Section "Evaluation of CPC profiles by CFD Simulation"), taking into account the heterogeneity of the bed by drawing the regular array of circles, which are considered dense and impermeable to gas (i.e., no slip condition at the particle wall). As a 2-D axial symmetry is used, the circles in 2-D becomes toroidal forms in 3-D. Therefore, the considered structure is actually composed of an array of rings, around which the gas mixture flows. In doing that, what is neglected in this system with respect to a real3-D one is the part of the flow around particles owing to the presence of the toroidal wall, which makes the cross-sectional area available to flow smaller than that offered by spherical particles.

Concerning the geometrical properties of the system, the considered membrane length is 20 cm with an outer diameter of 1 mm, whereas two different values of particle diameter are used to pack the shell (d_{Part} = 3.10 mm and 1.53 mm, respectively). In particular, in the construction of the considered 2D particle array, contact with particles is prevented, in order to avoid the retentate channel to be closed to the gas flow. In this way we want to analyze the effect of the perturbation induced by the presence of obstacles (particles) with respect to the empty channel case in a condition of low packing degree.

The values of the voidage degree (ε_{Bed}) – calculated to be 0.85 for the smaller-particle bed and 0.8 for the larger-particle one – are evaluated based on the 2D geometry, i.e., by the ratio of the void surface and the total one. Therefore, its value is actually different from the 3D one, remarking that these geometrical conditions, which are actually closer to fluidised beds ones, are considered just to describe practically the use of the *EAC* coefficient in determining the polarization extent from simulation data. At the same time, it is underlined that a deeper and more exhaustive analysis of this type would imply

Table 1 – Operating conditions and geometrical details of the simulated reactor.

Side	Composition, %				Total pressure	GHSV	T^{Inlet}
	H_2	CO_2	H_2O	CO	kPa	h^{-1}	°C
Feed	48.5	4.9	36.9	9.7	[400, 600, 800]	[20,000, 40,000]	[360, 400]
Sweep	100	–	–	–	[10, 20, 50]	No Sweep	Same as feed

ID^{Shell} = 1.4 cm, δ^{Shell} = 2 mm, OD^{Mem} = 1 cm, L^{Mem} = 6.25 cm, δ^{Mem} = 5 μm.

the effect of other parameters like types of arrays, particle shape and size, bed porosity and inlet velocity, which are beyond the scope of the present paper.

As for the simulation settings, the retentate and permeate pressure are fixed to 5 bar and 1 bar, respectively. Under these conditions, the evaluated average Reynolds number (Re) is 5.7 for both systems, whereas the particle Reynolds number (Re_p) is 0.70 for the bed with the larger particles ($\varepsilon_{Bed} = 0.850$) and 0.34 for that with the smaller ones ($\varepsilon_{Bed} = 0.825$).

The used boundary conditions for momentum transfer are: velocity field set at the inlet, total pressure set at the outlet and no-slip condition set on the membrane and particles surfaces.

As for the mass transfer equations, the used boundary conditions are: compositions set at the inlet, convective flux set at the outlet, conditions of impermeability on particle surface for all species (H_2 and N_2) and on membrane surface for N_2 only. The required hydrogen flux that has to be set on membrane surface is calculated by using the only internal diffusion step of the previously mentioned hydrogen permeation model, as the external mass transfer on the particle bed side is solved by the CFD solver.

As for CPC, its value at a certain reactor position is evaluated by first defining an appropriate characteristic permeation driving force (the Sieverts one) and then evaluating such a driving force both between the bulks of the gas phases and between the membrane surfaces on feed and permeate side. Concerning this point, we choose to take the hydrogen partial pressure of bulk as the maximum value along the radial direction at each position of reactor.

In fact, in general, owing to the presence of particles, the velocity and pressure field can be sufficiently complicated that it could be generally impossible to recognize a bulk value, as the definition of bulk would be questionable as well. Therefore, in this particular case, the bulk value of hydrogen partial pressure used to calculate *CPC* at a certain axial position z is evaluated by considering the maximum value of partial pressure at that position, whereas the membrane value is directly evaluated at the membrane surface. It is noticed that such a choice gives the highest value of the CPC.

Based on this approach, the explicit expression reported in Eq. (2) is used for *CPC* calculation.

$$CPC = 1 - \frac{\sqrt{P^{Feed}_{H_2,Surface}} - \sqrt{P^{Permeate}_{H_2,Surface}}}{\sqrt{P^{Feed}_{H_2,Bulk}} - \sqrt{P^{Permeate}_{H_2,Bulk}}} \quad (2)$$

This definition implies that *CPC* approaches zero for negligible polarization whereas approaches the unity for maximum one [2].

Results and discussion

Improved membrane characterisation in pure hydrogen conditions

As mentioned above, in this work the approach introduced by Hara and co-workers is used to evaluate the intrinsic permeability – referred to as *ideal permeability* – and the non-ideal contribution of the hydrogen internal diffusion [35,36,41].

Let us remark that the more standard characterization using the Sieverts law (Eq. (3)) does not take into account the permeability variation with hydrogen concentration, as a linear regression is used independently of the trend of the experimental data.

$$H_2Flux = H_2Permeance \times \left[\sqrt{P^{Feed}_{H_2}} - \sqrt{P^{Permeate}_{H_2}}\right] \quad (3)$$

In this sense, the regression results can be referred to as a sort of average with respect to the hydrogen concentration, considering the distance between data and fitting straight line a discrepancy due to just experimental error.

However, for increasing hydrogen concentration in the feed and/or permeate side, the discrepancy from the straight line is more and more caused by the effect of the non-ideal diffusion [35,41], whose behaviour implies a more complete membrane characterization.

As described elsewhere, the improved approach used in this work makes use of a third-order polynomial form without the second-order term to fit the experimental flux vs. the square root of the feed pressure at fixed permeate pressure [35,36,41]. As a result, the hydrogen permeability can be evaluated as a function of the hydrogen concentration (or, equivalently, pressure). As mentioned above, the experimental data reported in Brunetti et al. (2015) [31] are used to perform this analysis.

Fig. 3 shows this type of results at different temperatures. First of all, let us remind that this permeability is referred to as "*intrinsic*" because, for hydrogen pressure approaching to zero, the effect of internal stresses caused by the hydrogen permeation in the metal lattice is negligible and independent of hydrogen concentration. The values of the intrinsic permeability can be read in this type of plots at a hydrogen pressure approaching to zero, corresponding to a state of infinite dilution of the hydrogen atoms in the metal lattice.

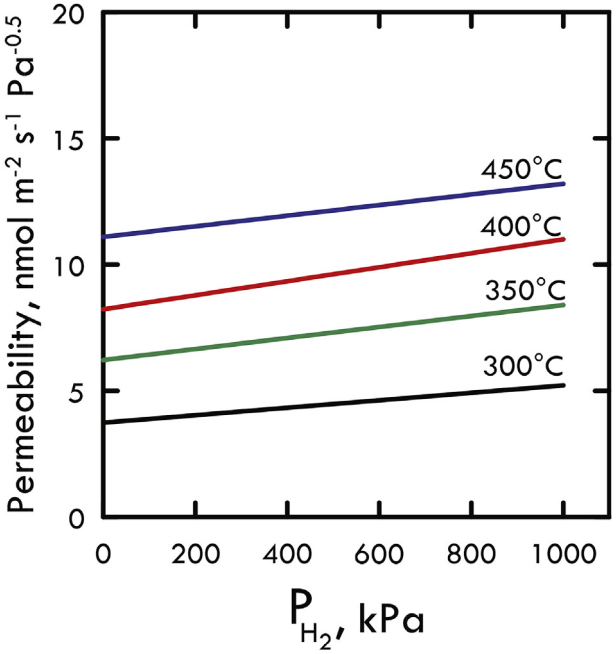


Fig. 3 – Hydrogen permeability as a function of hydrogen pressure calculated at different values of temperature.

As expected, it is shown that the intrinsic permeability is favoured by temperature, as it is an activated process that can be described in a first approximation by an Arrhenius-type expression (Eq. (4)) whose trend is shown in Fig. 4:

$$Pe_{Ideal} = Pe_0 \exp\left(-\frac{E_0}{RT}\right) \quad (4)$$

Concerning the functionality with the hydrogen pressure, the overall permeability is found to increase with it. From a physical point of view, this is because, within the pressure range considered in this investigation, the increased hydrogen concentration in the metal lattice augments the lattice–hydrogen interactions, causing stress relaxations favouring permeation.

A quantification of this effect can be provided by considering a further evolution of this approach, allowing the non-ideal contribution of the internal diffusion to be evaluated from the deviation from ideality by splitting the polynomial flux expression into ideal and non-ideal terms [41]. The result of this analysis is shown in Fig. 5, where the non-ideality index is depicted as a function of the hydrogen feed pressure at different temperatures.

In this figure, it is found that such an index – and, thus, the system non-ideality – increases with increasing hydrogen pressure and with decreasing temperature. The functionality with pressure is caused by the same reasons as those commented above, whereas the observed functionality with temperature is basically due to the fact that a higher temperature decreases the lattice–hydrogen interactions, this pushing the system towards a more ideal behaviour.

Furthermore, this trend with temperature indicates that the influence of the porous support on permeation is negligible. In fact, if support had an appreciable influence, the apparent non-ideality index would be increasing with temperature.

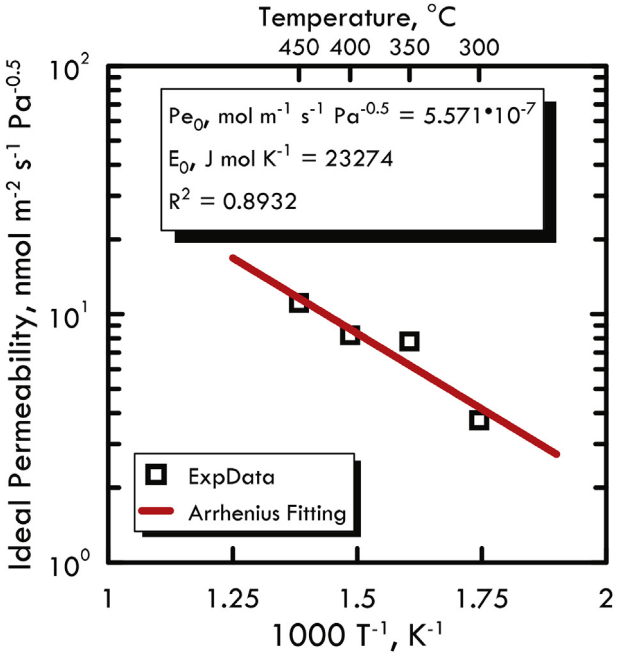


Fig. 4 – Arrhenius-type plot of the ideal permeability of the considered membrane.

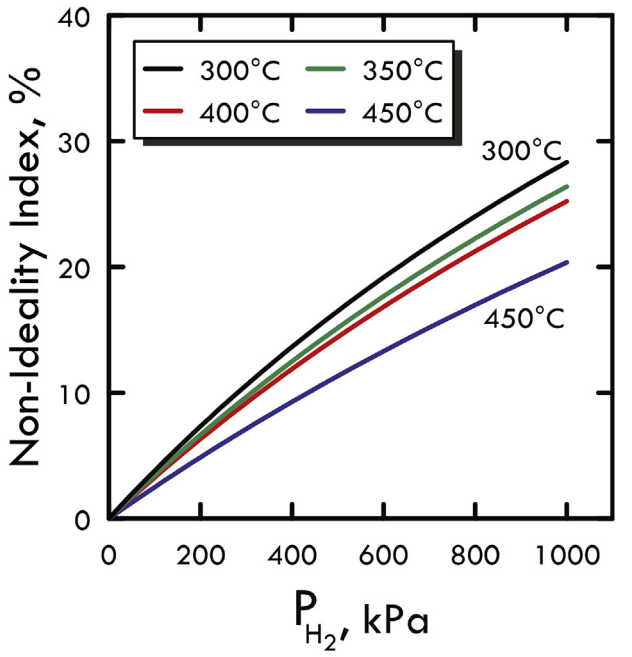


Fig. 5 – Ideality Index as a function of hydrogen pressure calculated at different values of temperature.

However, in that case, this index would not be representative of the non-ideality of internal diffusion – that is why would be named as "*apparent*" – and, thus, its values would indicate that the influence of the support should be separated from that of the membrane to correctly evaluate the membrane permeability.

Evaluation of CPC profiles by CFD simulation

As described above, the permeability obtained in the previous section is used to simulate the system described above (Section "CFD Simulation for Concentration Polarization") by CFD technique. Fig. 6b–c depict the calculated velocity profile, showing the role of particles in increasing the local velocity field between two consecutive surfaces. This effect makes the local Reynolds between particles and membrane surface higher with respect to a system without particles, which are expected then to enhance the external mass transfer towards membrane and, thus, to decrease the *CPC* value.

As mentioned above, the axial *CPC* profiles along the particle bed are obtained from the calculated 2-D hydrogen concentration profile at each position of the system.

The obtained results are shown in Fig. 7 along with the axial hydrogen partial pressure profiles, which are calculated by taking the average value along the radial direction at each axial position. In all configurations, the pressure drop along reactor is found to be negligible.

Additionally, two types of average *CPC* are shown, i.e., the standard average *CPC* (CPC_{Ave}, Eq. (5)) and the *Effective Average CPC* (*EAC*, Eq. (6)), the latter of which was recently introduced and used by Caravella and Sun (2015) to provide a more correct evaluation of the overall polarization level in the system avoiding undesired over-estimations [42].

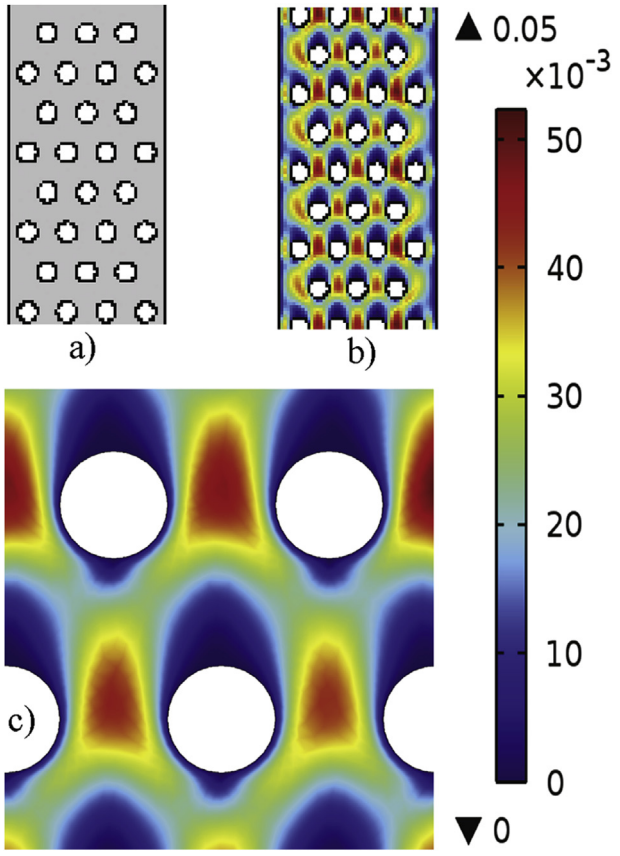


Fig. 6 – Detail of the array of (a) spherical particles used to simulate the concentration polarization in bed, (b) inter-particle velocity profile and (c) a zoom of it.

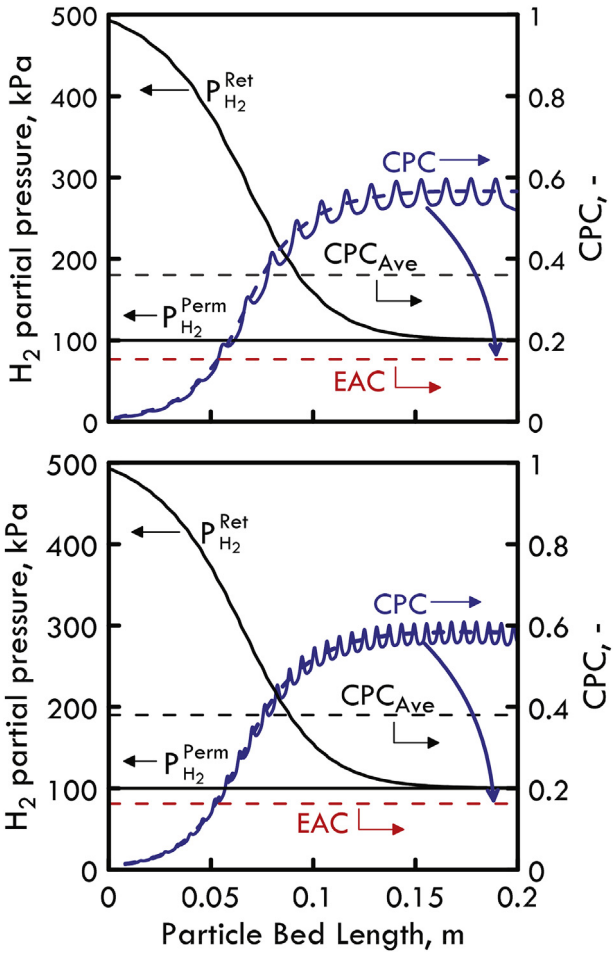


Fig. 7 – CPC and hydrogen partial pressure profiles in the simulated particle beds along with the standard average CPC_{Ave} and EAC. a) Larger and b) smaller particles. The blue curved arrows indicate the overall level of polarization in the particle bed, given by the EAC coefficient. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$CPC_{Ave} = \overline{CPC} \equiv \frac{1}{L}\int_0^L CPC(z)dz \tag{5}$$

$$EAC = 1 - \frac{[\text{Permeation Rate with Polarization}]}{[\text{Permeation Rate without Polarization}]} = 1 - \frac{\int_0^L Flux(z)dz}{\int_0^L \frac{Flux(z)}{1-CPC(z)}dz} \tag{6}$$

In fact, overall speaking, concentration polarization affects the reactor performance. A measure of such a performance loss can be given by both CPC_{Ave} and EAC, which are post-processing parameters and, thus, their values do not affect the reactor performances. As mentioned above, their calculation can be performed as long as the CPC profile along reactor is available. Usually this is made indirectly, which means that first a model able to well describe the reactor performance has to be developed, and then both CPC_{Ave} and EAC are calculated from the simulated CPC profile.

The need of using EAC – extensively reported in Ref. [42] – can be clearly understood by looking at the shape of the CPC profiles. In fact, considering Fig. 7a (larger particles), starting from the begin of reactor, it is observed that the CPC assumes a value close to zero, owing to the relatively high hydrogen concentration at the inlet. More specifically, the polarization over-estimation can be simply measured as the difference between them, remarking that, by definition, EAC is always lower than CPC_{Ave}.

Going towards the end of reactor, CPC becomes progressively higher because of the higher concentration of nitrogen owing to the hydrogen permeation through membrane. However, the permeating flux in this last reactor section is smaller, accounting less, and less on CPC. This gives an EAC value of ca. 0.18 much lower of the local CPC value of ca. 0.6.

The saw-tooth trend is because the presence of the regular array of particles causes the hydrogen partial pressure to oscillate with the same regularity. It is also interesting to notice that the saw-tooth shape is less pronounced at the inlet.

This is because the hydrogen partial pressure change due to the change of the total pressure field among particles is less influenced by the particle array owing to the hydrogen content

close to the unity. However, we can consider that the actual CPC profile is that identified by the average value among the waves, indicated by the blue dashed-line in the plot. Considering Fig. 7b (smaller particles), a similar trend is observed, except for the saw-tooth profile, which is clearly narrower owing to the smaller particle size. The fact that the average CPC profile relative to the larger particles is found to be slightly lower than that relative to the smaller particles – this causing both CPC_{Ave} and EAC to be lower in the former case – is due to the smoother distribution of the velocity field as the particle size becomes progressively lower, tending to a radial uniform distribution in the limit case of infinitely small particles. An EAC value (<0.2) much lower of the local CPC one (ca. 0.6) can be observed in the case of smaller particles.

As for the CPC_{Ave} and EAC, the former provides an overestimation of the overall polarization level in the reactor because its value increases in the neighbourhood of a null driving force, which, however, does not give any appreciable contribution to the reactor performance.

On the contrary, the latter does not take into account the zones where the driving force is small and, thus represent a more realistic parameter to measure the polarization in membrane-assisted devices. Next section provides this type of analysis applied to the 1D–1D simulation of the considered WGS membrane reactor.

Simulation of membrane reactor for WGS

Before simulating WGS in a Pd-based membrane reactor, the reactor model including concentration polarization is validated using literature data [31] in terms of CO conversion (Fig. 8a) and H_2 Recovery Index (Fig. 8b) vs. total feed pressure. The observed good agreement between experimental and simulated data, indicates the model efficacy to reproduce the actual reactor behaviour.

Such a good agreement is established without any adjustable parameter. The only arbitrary choice for parameters is the catalyst intra-particle tortuosity, supposed to vary within a range of 2–7. However, the simulation sensitivity to this geometrical property is observed to be low and, thus, a tortuosity value of 3 is used in all the subsequent calculation.

Furthermore, it is found that the conversion is slightly increasing with total pressure (actually almost constant), whereas the recovery index increases significantly with it. This indicates that, under the conditions considered in the reaction tests, the kinetics is weakly influenced by the presence of the membrane, which basically acts to increase the pure hydrogen recovery extent in the permeate. In fact, the WGS kinetics is thermodynamically not influenced by the total pressure and the positive effect of the pressure is due to the membrane, which allows a continuous hydrogen removal pushing the equilibrium towards the products.

Therefore, the validation of the used concentration polarization model is more effective in this case if made in terms of hydrogen recovery, which is much more sensitive to the polarization influence. After validation, the system described in Fig. 2 is simulated under the operating conditions reported in Table 1. Once more, it must be specified that we do not have a single value of CPC along reactor, as the concentration of the species involved in the process changes as the reaction proceeds towards reactor outlet.

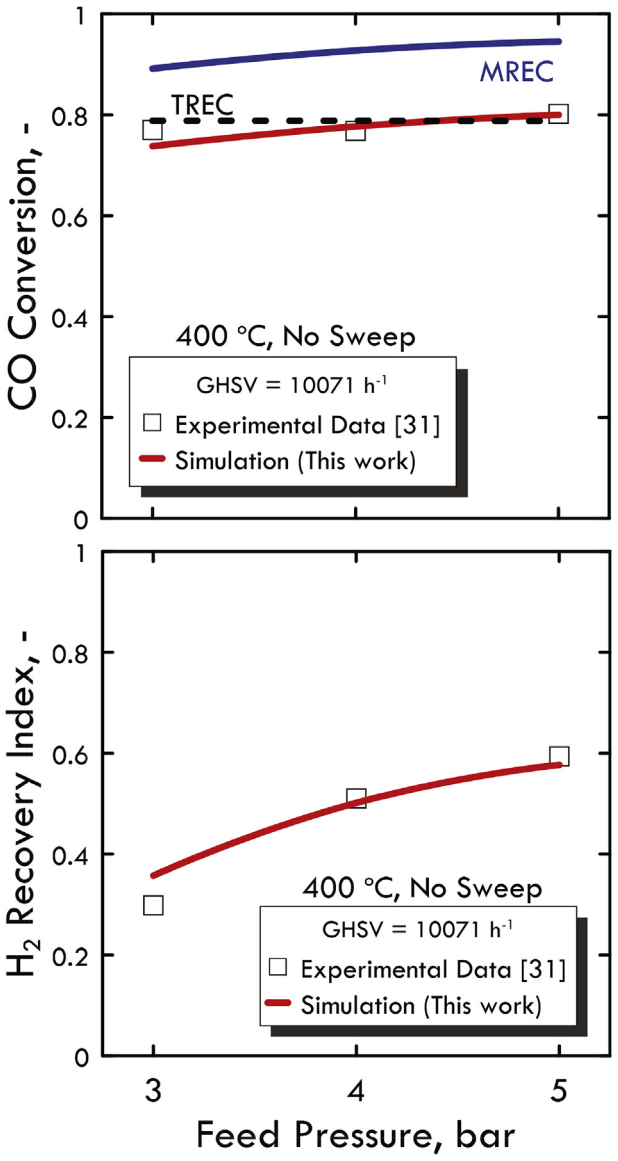


Fig. 8 – CO conversion (a) and H_2 Recovery Index (b) as a function of the Sieverts driving force at an inlet temperature of 400 °C: experimental data (symbols) [31] and simulation (solid line). The curves of both the corresponding Membrane Reactor Equilibrium Conversion (MREC, blue solid line) and the Traditional Reactor Equilibrium Conversion (TREC, black dashed-line) is also plotted. Composition: H_2:CO_2:H_2O:CO = 48.53:4.94:36.87:9.66. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Figs. 9–12 show some simulation results depicting profiles of hydrogen partial pressures and CPC along with CPC_{Ave} and EAC (Eqs. (5) and (6)). Fig. 9 shows some profiles calculated at a total feed pressure of 800 kPa, inlet temperature of 360 °C and GHSV of 40,000 h^{-1}. As for the hydrogen profile, it first increases owing to the fact that the reaction rate is faster than

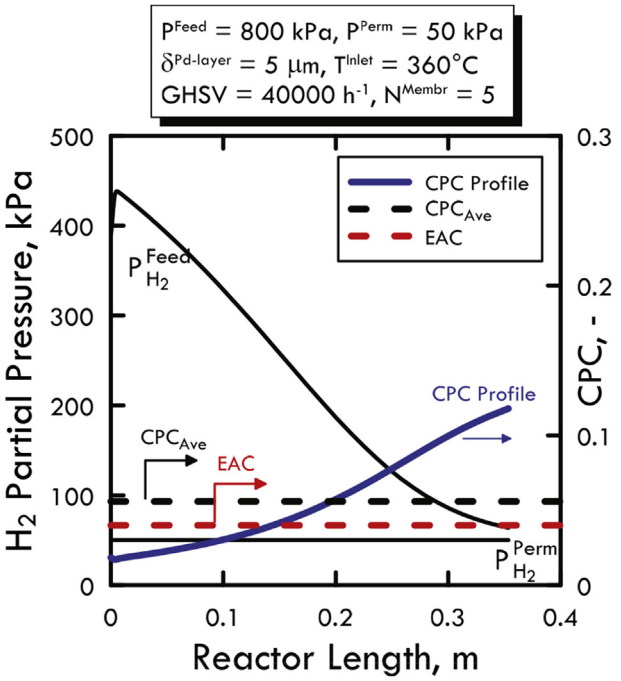


Fig. 9 – Hydrogen partial pressure profiles (black solid line), concentration polarization coefficient profile (blue solid line) along the WGS reactor and average values (both CPC_{Ave} and EAC, dashed black and red lines, respectively) evaluated at the conditions reported in the legend. Average Re = 36.9. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

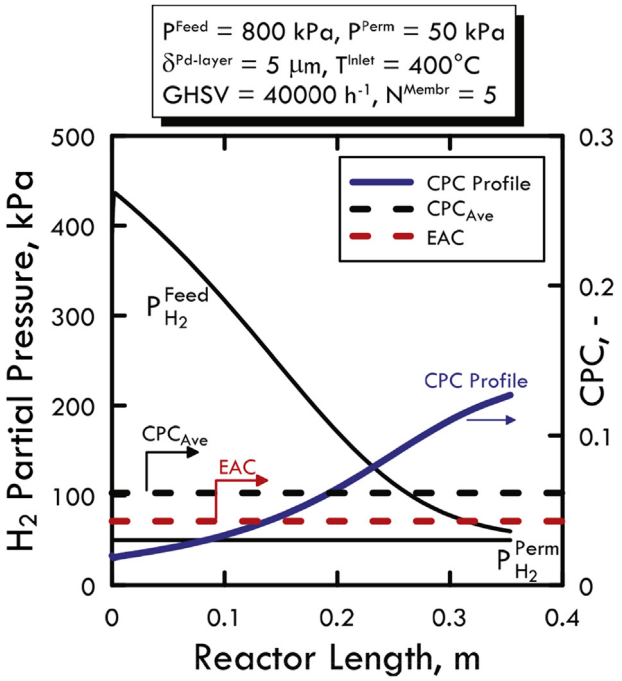


Fig. 10 – Hydrogen partial pressure profiles (black solid line), concentration polarization coefficient profile (blue solid line) along the WGS reactor and average values (both CPC_{Ave} and EAC, dashed black and red lines, respectively) evaluated at the conditions reported in the legend. Average Re = 35.4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the permeation one at the beginning of reactor, whereas the contrary occurs towards the end.

As for the concentration polarization one, it follows almost the same trend as the hydrogen partial pressure, i.e., first slightly decreasing and then increasing continuously up to the reactor end. This trend strongly depends on the concentration of the non-permeating species in the reaction side, which increases owing to the hydrogen permeation through membrane.

Because of the relatively high value of GHSV considered, the fluid dynamics is favoured and, thus, both CPC_{Ave} and EAC values are quite small (around 0.050 and 0.035), this indicating that concentration polarization is basically negligible in these conditions. Furthermore, the similarity of CPC_{Ave} and EAC indicates that the extent of the reactor zone with small driving force is small as well.

Considering a higher inlet temperature (400 °C, Fig. 10), shows similar trends but higher polarization value caused by the higher concentration of non-permeating species owing to the higher transmembrane flux favoured by the higher temperature. The CPC_{Ave} and EAC values – 0.062 and 0.043, respectively – quantitatively indicate such a polarization increase, even though the Reynolds number is slightly higher.

Additionally, the larger difference among them indicates that the region with a relatively small driving force is larger, this being in line with the fact that, for higher permeation rate, the system tends towards the equilibrium faster (i.e., towards the reactor outlet) at the same GHSV.

Considering a lower total feed pressure than the previous case (600 kPa) by keeping the same other conditions (Fig. 11), it is possible to observe a lower polarization level (with CPC_{Ave} and EAC values of 0.041 and 0.033, respectively). This occurs because at lower total pressure, the permeating flux is lower due to a lower driving force.

However, one of the most critical operating conditions for reactor is the GHSV, not only from the productivity point of view, but also from the fluid dynamics one, which crucially affects the polarization level. In fact, considering the same conditions as those used in Fig. 10 but decreasing the GHSV value to 20,000 h^{-1} (one and a half value), a quite different behaviour is observed (Fig. 12). In fact, the partial pressure profiles show that the reactor is close to the equilibrium at around 0.2 cm from the inlet, from which on the driving force is practically null. In this case, the CPC_{Ave} and EAC values are quite different (i.e., 0.142 and 0.066, respectively), this stating that the polarization level in the zone approaching the equilibrium (i.e., from around 0.2 on) is significant. However, as said before, this zone does not influence the reactor performance and, thus, the value of 0.142 is just an over-estimation.

An interesting aspect to notice is that a decreased GHSV not only causes the reactor to be under-exploited from a point of view of catalyst and membrane area used, but also causes the fluid dynamics to disfavour the concentration polarization

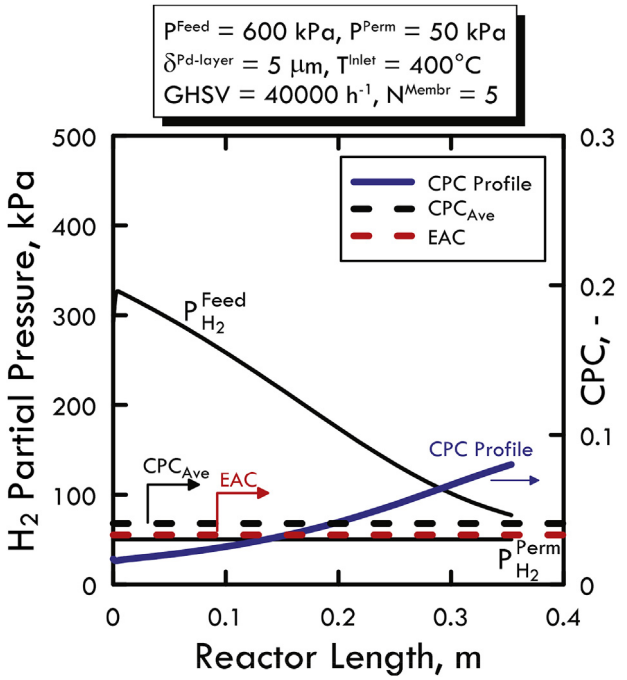


Fig. 11 – Hydrogen partial pressure profiles (black solid line), concentration polarization coefficient profile (blue solid line) along the WGS reactor and average values (both CPC_{Ave} and EAC, dashed black and red lines, respectively) evaluated at the conditions reported in the legend. Average $Re = 35.9$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

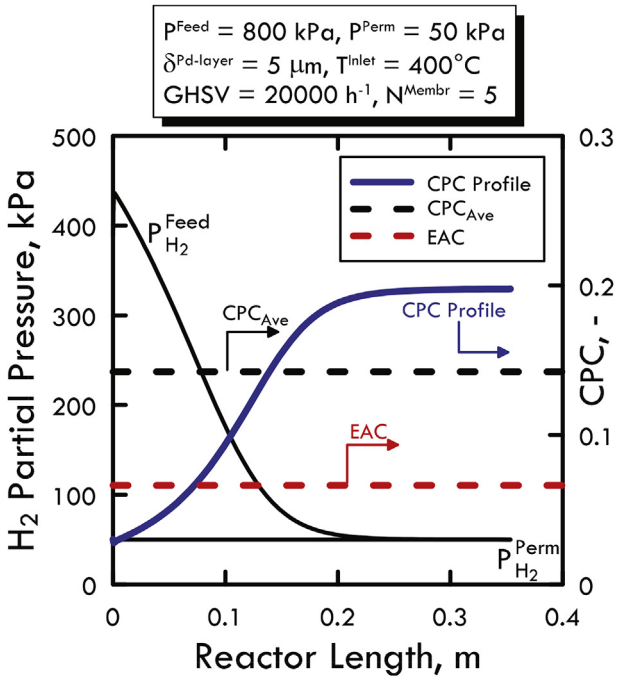


Fig. 12 – Hydrogen partial pressure profiles (black solid line), concentration polarization coefficient profile (blue solid line) along the WGS reactor and average values (both CPC_{Ave} and EAC, dashed black and red lines, respectively) evaluated at the conditions reported in the legend. Average $Re = 17.1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in the reactor, as the mass transfer film thickness increases and, correspondingly, the mass transfer coefficient decreases. This is important not only for mass transfer towards membrane, but also towards the catalyst surface, enhancing the effectiveness factor and, thus, the overall reactor performance.

Conclusions

In this work, the concentration polarization distribution in Pd-based membrane reactors for Water Gas shift was investigated by a simulation approach coupling an improved characterization of a 3.6 μm-thick membrane using permeation data from the literature, CFD simulations of particle beds and a complex model of membrane reactor.

First, a real 3.6 μm-thick membrane by an improved approach allowing the determination of both hydrogen permeability and non-ideality index of hydrogen internal diffusion as a function of temperature and hydrogen pressure.

Then, the flow of a binary mixture of H_2/N_2 through a particle bed of a regular 2D arrays of mono-disperse spherical particles was analyzed by computational fluid dynamics simulations. This analysis showed that the velocity field between particles and membranes contributes to enhance the mass transfer towards the membrane surface and that the particle size does not provide an appreciable contribution in changing the concentration polarization level in the reactor, at least for mono-disperse particles. In the two cases analyzed, an overall polarization level of around 18% was calculated.

Finally, the used polarization model was indirectly validated by experimental data from the literature [31] in terms of CO conversion and H_2 Recovery Index vs. feed pressure, finding a satisfactory agreement. Afterwards, further simulations of the membrane reactor with five membranes were carried out. This allowed the overall polarization level to be calculated in terms of both the standard average CPC (CPC_{Ave}) and the *Effective Average CPC* (EAC), the latter of which avoid over-estimation of the polarization level in the reactor zones with a small driving force.

In this sense, the knowledge of CPC profile along membrane reactors provides useful information to recognize those reactor zones where polarization mostly affects the overall performance and, thus, allows optimal catalyst and/or membrane distributions to be adopted.

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Notation

d	Diameter (particle), m
L	Reactor Length, m
k_{CO}	WGS Kinetic constant, mol kg^{-1} s^{-1} Pa^{-2}
K_{Eq}	WGS Equilibrium constant, –
P	Pressure, Pa
T	Temperature, K
z	Reactor abscissa, m

Subscripts/Superscripts

Ave	Average
i	*i*-th species in mixture
Mem	Membrane
Part	Particle (in the particle bed)
Perm	Permeate
Ret	Retentate

Greek letters

δ	Thickness (membrane and shell annulus), m
ϵ_{Bed}	Voidage Degree, –
τ_{Bed}	Tortuosity, –

Acronyms/Abbreviations

CPC	Concentration Polarization Coefficient
EAC	Effective Average CPC
CFD	Computational Fluid Dynamics
GHSV	Gas Hourly Space Velocity, h^{-1}
ID	Inner Diameter, m
OD	Outer Diameter, m
Perm	Permeate
Ret	Retentate

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